Aws credentials

```
ubuntu@masternode:~$ aws configure
AWS Access Key ID [None]: AAAAAAAAAAAA
AWS Secret Access Key [None]: AAAAAABBBCCC+
Default region name [None]: eu-north-1
Default output format [None]:
```

Create a cluster

```
ubuntu@masternode:~$ eksctl create cluster --name project-eks --region eu-north-1 --version 1.33 --
node-type t3.medium --nodes 2           ## may take time up to 15 mins
## minimum t3.medium for kubernetes
```

// OR

```
ubuntu@masternode:~$ eksctl create cluster \
  --name project-eks \
  --region eu-north-1 \
  --version 1.33 \
  --nodegroup-name project-nodegroup \
  --node-type t3.medium \
  --nodes 2 \
  --nodes-min 1 \
  --nodes-max 3
```

Validation

```
ubuntu@masternode:~$ kubectl config get-contexts
CURRENT   NAME                                          CLUSTER                              AUTHINFO
NAMESPACE
*         eks-console@project-eks.eu-north-1.eksctl.io  project-

ubuntu@masternode:~$ kubectl get nodes
NAME                                          STATUS   ROLES    AGE    VERSION
ip-192-168-25-78.eu-north-1.compute.internal  Ready    <none>   2m7s   v1.33.0-eks-802817d
ip-192-168-32-147.eu-north-1.compute.internal Ready    <none>   2m6s   v1.33.0-eks-802817d
```

Scalabilty Achieved

```
ubuntu@masternode:~$ kubectl get nodes
NAME                                          STATUS   ROLES    AGE     VERSION
ip-192-168-0-240.eu-north-1.compute.internal  Ready    <none>   4m15s   v1.33.0-eks-802817d
ip-192-168-37-32.eu-north-1.compute.internal  Ready    <none>   14s     v1.33.0-eks-802817d



## Make sure you did add policy for ebs and efs
## After add policy run this commands for more ensure


ubuntu@masternode:~$ kubectl apply -k "github.com/kubernetes-sigs/aws-ebs-
csi-driver/deploy/kubernetes/overlays/stable/?ref=release-1.14"

ubuntu@masternode:~$ kubectl get csidrivers
NAME             ATTACHREQUIRED   PODINFOONMOUNT   STORAGECAPACITY   TOKENREQUESTS   REQUIRESREPUBLISH   MODES        AGE
ebs.csi.aws.com  true             false            false             <unset>         false               Persistent   5s
efs.csi.aws.com  false            false            false             <unset>         false               Persistent   12m

ubuntu@masternode:~$ kubectl apply -k
"github.com/kubernetes-sigs/aws-efs-csi-driver/deploy/kubernetes/overlays/stable/?ref=release-1.5"

ubuntu@masternode:~$ kubectl get csinodes
NAME                                           DRIVERS   AGE
ip-192-168-30-15.eu-north-1.compute.internal   2         17m
ip-192-168-43-226.eu-north-1.compute.internal  2         17m
```

```
ubuntu@masternode:~$ mkdir deploysite-project
ubuntu@masternode:~$ cd deploysite-project/
ubuntu@masternode:~/deploysite-project$ echo -n 'putyourpassword' | openssl base64
bGFtb25hb2Z0aGlzZW1vYXNl
```

mysql-secret.yaml

```
ubuntu@masternode:~/deploysite-project$ vim mysql-secret.yaml

ubuntu@masternode:~/deploysite-project$ kubectl apply -f mysql-secret.yaml
secret/secure-pass created

ubuntu@masternode:~/deploysite-project$ kubectl get secret
NAME          TYPE     DATA   AGE
secure-pass   Opaque   1      9s


ubuntu@masternode:~/deploysite-project$ kubectl describe secret
Name:         secure-pass
Namespace:    default
Labels:       <none>
Annotations:  <none>

Type:  Opaque

Data
====
password:  18 bytes
```

mysql-storageclass.yaml

```
ubuntu@masternode:~/deploysite-project$ vim mysql-sc.yaml

ubuntu@masternode:~/deploysite-project$ kubectl apply -f mysql-sc.yaml
storageclass.storage.k8s.io/mysql-sc created

ubuntu@masternode:~/deploysite-project$ kubectl get sc


ubuntu@masternode:~/deploysite-project$ kubectl get sc
NAME       PROVISIONER            RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
mysql-sc   ebs.csi.aws.com        Delete          WaitForFirstConsumer   false                  10s

ubuntu@masternode:~/deploysite-project$ kubectl describe sc mysql-sc
Name:            mysql-sc
IsDefaultClass:  No
Annotations:     kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/
v1","kind":"StorageClass","metadata":{"annotations":{},"name":"mysql-
sc"},"provisioner":"ebs.csi.aws.com","volumeBindingMode":"WaitForFirstConsumer"}

Provisioner:           ebs.csi.aws.com
Parameters:            <none>
AllowVolumeExpansion:  <unset>
MountOptions:          <none>
ReclaimPolicy:         Delete
VolumeBindingMode:     WaitForFirstConsumer
Events:                <none>
```

mysql-pvc.yaml

```
ubuntu@masternode:~/deploysite-project$ vim mysql-pvc.yaml

ubuntu@masternode:~/deploysite-project$ kubectl apply -f mysql-pvc.yaml
persistentvolumeclaim/mysql-pvc created

ubuntu@masternode:~/deploysite-project$ kubectl get pvc
NAME        STATUS    VOLUME   CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mysql-pvc   Pending                                      mysql-sc       <unset>                 35s


ubuntu@masternode:~/deploysite-project$ kubectl get pvc
kuNAME        STATUS    VOLUME   CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mysql-pvc   Pending                                      mysql-sc       <unset>                 5s

ubuntu@masternode:~/deploysite-project$ kubectl describe pvc
Name:          mysql-pvc
Namespace:     default
StorageClass:  mysql-sc
Status:        Pending
Volume:
Labels:        <none>
Annotations:   <none>
Finalizers:    [kubernetes.io/pvc-protection]
Capacity:
Access Modes:
VolumeMode:    Filesystem
Used By:       <none>
Events:
  Type    Reason                Age                From                         Message
  ----    ------                ----               ----                         -------
  Normal  WaitForFirstConsumer  1s (x2 over 10s)  persistentvolume-controller  waiting for first
consumer to be created before binding
```

mysql-deploy.yaml

```
ubuntu@masternode:~/deploysite-project$ vim mysql-deploy.yaml

ubuntu@masternode:~/deploysite-project$ kubectl apply -f mysql-deploy.yaml
deployment.apps/mysql-deployment created

ubuntu@masternode:~/deploysite-project$ kubectl get deploy
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
mysql-deployment   1/1     1            1           28s

ubuntu@masternode:~/deploysite-project$ kubectl get po
NAME                                READY   STATUS    RESTARTS   AGE
mysql-deployment-5bbcfd489b-t8bkl   1/1     Running   0          30s

ubuntu@masternode:~/deploysite-project$ kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
mysql-deployment-5bbcfd489b   1         1         1       34s
```

Validation

```
ubuntu@masternode:~/deploysite-project$ kubectl get pv
kuNAME                                       CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM               STORAGECLASS
VOLUMEATTRIBUTESCLASS   REASON   AGE
pvc-9c7f3863-f216-4636-b886-d587a72a1a10   5Gi        RWO            Delete           Bound    default/mysql-pvc   mysql-sc
<unset>                          39s

ubuntu@masternode:~/deploysite-project$ kubectl get pvc
NAME        STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mysql-pvc   Bound    pvc-9c7f3863-f216-4636-b886-d587a72a1a10   5Gi        RWO            mysql-sc       <unset>
2m31s

ubuntu@masternode:~/deploysite-project$ kubectl get sc
NAME       PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
gp2        kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer   false                  38m
mysql-sc   ebs.csi.aws.com         Delete          WaitForFirstConsumer   false                  6m58s
```

☐	vol-03fbacd80c29d934a	gp3	5 GiB	3000	125	-	-

mysql-service.yaml

```
ubuntu@masternode:~/deploysite-project$ vim mysql-service.yaml

ubuntu@masternode:~/deploysite-project$ kubectl apply -f mysql-service.yaml
service/mysql-svc created

ubuntu@masternode:~/deploysite-project$ kubectl get svc
NAME         TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
kubernetes   ClusterIP   10.100.0.1       <none>        443/TCP    39m
mysql-svc    ClusterIP   10.100.177.176   <none>        3306/TCP   24s

ubuntu@masternode:~/deploysite-project$ kubectl describe svc mysql-svc
Name:              mysql-svc
Namespace:         default
Labels:            <none>
Annotations:       <none>
Selector:          app=wordpress,tier=mysql
Type:              ClusterIP
IP Family Policy:  SingleStack
IP Families:       IPv4
IP:                10.100.177.176
IPs:               10.100.177.176
Port:              <unset>  3306/TCP
TargetPort:        3306/TCP
Endpoints:         192.168.43.107:3306
Session Affinity:  None
Events:            <none>
```

site-sc.yaml

```
ubuntu@masternode:~/deploysite-project$ vim site-sc.yaml

ubuntu@masternode:~/deploysite-project$ kubectl apply -f site-sc.yaml
storageclass.storage.k8s.io/site-sc created

ubuntu@masternode:~/deploysite-project$ kubectl get sc
NAME       PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
gp2        kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer   false                  56m
mysql-sc   ebs.csi.aws.com         Delete          WaitForFirstConsumer   false                  17m
site-sc    efs.csi.aws.com         Delete          Immediate              false                  7s
```

Create AWS EFS Using console with Access Point

site-pv.yaml

```
ubuntu@masternode:~/deploysite-project$ vim site-pv.yaml

ubuntu@masternode:~/deploysite-project$ kubectl apply -f site-pv.yaml
persistentvolume/site-pv created
u
buntu@masternode:~/deploysite-project$ kubectl get pv
NAME                                       CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS      CLAIM               STORAGECLASS   VOLUMEATTRIBUTESCLASS   REASON   AGE
pvc-9c7f3863-f216-4636-b886-d587a72a1a10   5Gi        RWO            Delete           Bound       default/mysql-pvc   mysql-sc       <unset>                          14m
site-pv                                    5Gi        RWX            Retain           Available                       site-sc        <unset>                          6s


ubuntu@masternode:~/deploysite-project$ kubectl describe pv
Source:
    Type:              CSI (a Container Storage Interface (CSI) volume source)
    Driver:            efs.csi.aws.com
    FSType:
    VolumeHandle:      fs-04573f02800455565::fsap-009750f838ef2652b
    ReadOnly:          false
    VolumeAttributes:  <none>
Events:                <none>
```

site-pvc.yaml

```
ubuntu@masternode:~/deploysite-project$ vim site-pvc.yaml

ubuntu@masternode:~/deploysite-project$ kubectl apply -f site-pvc.yaml
persistentvolumeclaim/site-pvc created

ubuntu@masternode:~/deploysite-project$ kubectl get pvc
NAME        STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
mysql-pvc   Bound    pvc-9c7f3863-f216-4636-b886-d587a72a1a10   5Gi        RWO            mysql-sc       <unset>                 22m
site-pvc    Bound    site-pv                                    5Gi        RWX            site-sc        <unset>                 6s


ubuntu@masternode:~/deploysite-project$ kubectl get pv
NAME                                       CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM               STORAGECLASS   VOLUMEATTRIBUTESCLASS   REASON   AGE
pvc-9c7f3863-f216-4636-b886-d587a72a1a10   5Gi        RWO            Delete           Bound    default/mysql-pvc   mysql-sc       <unset>                          20m
site-pv                                    5Gi        RWX            Retain           Bound    default/site-pvc    site-sc        <unset>                          5m56s
```

site-deploy.yaml

```
ubuntu@masternode:~/deploysite-project$ aws efs describe-mount-targets --file-system-id
fs-04573f02800455565
{
    "MountTargets": [
        {
            "OwnerId": "311151994102",
            "MountTargetId": "fsmt-01e6d9446824db576",
            "FileSystemId": "fs-04573f02800455565",
            "SubnetId": "subnet-057eb3abf03a8c092",
            "LifeCycleState": "available",
            "IpAddress": "192.168.60.140",
            "NetworkInterfaceId": "eni-0fe163bd8a48b217c",
            "AvailabilityZoneId": "eun1-az2",
            "AvailabilityZoneName": "eu-north-1b",
            "VpcId": "vpc-0f02989a78f647468"
        },
        {
            "OwnerId": "311151994102",
            "MountTargetId": "fsmt-08495997dd2f6773b",
            "FileSystemId": "fs-04573f02800455565",
            "SubnetId": "subnet-0c7886590dfe22797",
            "LifeCycleState": "available",
            "IpAddress": "192.168.7.45",
            "NetworkInterfaceId": "eni-05b99b86d1953e743",
            "AvailabilityZoneId": "eun1-az1",
            "AvailabilityZoneName": "eu-north-1a",
            "VpcId": "vpc-0f02989a78f647468"


ubuntu@masternode:~/deploysite-project$ vim site-deploy.yaml

ubuntu@masternode:~/deploysite-project$ kubectl apply -f  site-deploy.yaml

ubuntu@masternode:~/deploysite-project$ kubectl get po
NAME                                  READY   STATUS    RESTARTS   AGE
mysql-deployment-5bbcfd489b-t8bkl     1/1     Running   0          47m
wordpress-deployment-bdfb5cc4-zln5b   1/1     Running   0          8m28s

ubuntu@masternode:~/deploysite-project$ kubectl get rs
kNAME                            DESIRED   CURRENT   READY   AGE
mysql-deployment-5bbcfd489b      1         1         1       55m
wordpress-deployment-bdfb5cc4    1         1         1       17m


ubuntu@masternode:~/deploysite-project$ kubectl get deploy
NAME                   READY   UP-TO-DATE   AVAILABLE   AGE
mysql-deployment       1/1     1            1           56m
wordpress-deployment   1/1     1            1           17m
```

site-svc.yaml

```
ubuntu@masternode:~/deploysite-project$ kubectl get svc
NAME         TYPE           CLUSTER-IP       EXTERNAL-IP                                                               PORT(S)
AGE
kubernetes   ClusterIP      10.100.0.1       <none>                                                                    443/TCP
121m
mysql-svc    ClusterIP      10.100.177.176   <none>                                                                    3306/TCP
82m
site-svc     LoadBalancer   10.100.113.122   a32f0d223963f4ce5acbdb613129dcbe-1349467392.eu-north-1.elb.amazonaws.com   80:31306/TCP
2m6s
```

validation

```
ubuntu@masternode:~/deploysite-project$ kubectl get svc
NAME         TYPE           CLUSTER-IP       EXTERNAL-IP                                                               PORT(S)
AGE
kubernetes   ClusterIP      10.100.0.1       <none>                                                                    443/TCP
132m
mysql-svc    ClusterIP      10.100.177.176   <none>                                                                    3306/TCP
93m
site-svc     LoadBalancer   10.100.113.122   a32f0d223963f4ce5acbdb613129dcbe-1349467392.eu-north-1.elb.amazonaws.com   80:31306/TCP
12m
```

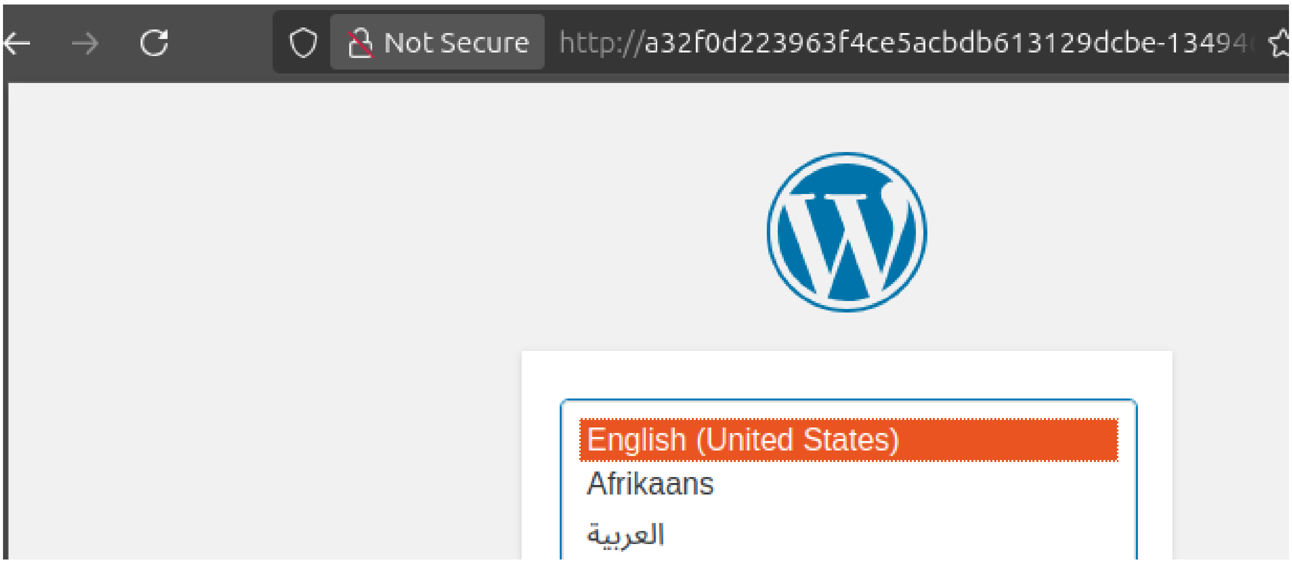
```
ubuntu@masternode:~/deploysite-project$ kubectl get nodes -o wide
NAME                                            STATUS   ROLES    AGE    VERSION               INTERNAL-IP      EXTERNAL-IP     OS-
IMAGE                      KERNEL-VERSION                  CONTAINER-RUNTIME
ip-192-168-30-15.eu-north-1.compute.internal    Ready    <none>   126m   v1.33.0-eks-802817d   192.168.30.15    13.61.144.215
Amazon Linux 2023.8.20250707   6.1.141-165.249.amzn2023.x86_64   containerd://1.7.27
ip-192-168-43-226.eu-north-1.compute.internal   Ready    <none>   126m   v1.33.0-eks-802817d   192.168.43.226   13.48.136.150
Amazon Linux 2023.8.20250707   6.1.141-165.249.amzn2023.x86_64   containerd://1.7.27
```

```
### Checking website using publicip:nodeport
### Make sure security group of your ec2 has an open port custom tcp : 31306
### Elb will be created Auto with all sg needed no need to edit
```

13.61.144.215:31306

Checking using the AWS ELB DNS

DNS: a32f0d223963f4ce5acbdb613129dcbe-1349467392.eu-north-1.elb.amazonaws.com



Clean

```
ubuntu@masternode:~$ kubectl delete -f deploysite-project
secret "secure-pass" deleted
deployment.apps "mysql-deployment" deleted
persistentvolumeclaim "mysql-pvc" deleted
storageclass.storage.k8s.io "mysql-sc" deleted
service "mysql-svc" deleted
deployment.apps "wordpress-deployment" deleted
persistentvolume "site-pv" deleted
persistentvolumeclaim "site-pvc" deleted
storageclass.storage.k8s.io "site-sc" deleted
service "site-svc" deleted
```

```
ubuntu@masternode:~$ eksctl delete cluster --name project-awseks
```